Customer Segmentation & Prediction

Week 11 Capstone: Business Recommendations

Dataset: 500 records and 9 columns. Target: Churn (0/1). CustomerID excluded from features.

Methodology

Numeric features were median-imputed and standardized; categorical features were mode-imputed and one-hot encoded. K-Means produces three operational segments. Hierarchical clustering and DBSCAN are compared. Separate Random Forest classifiers are tuned within each segment where sample size and class counts permit.

Segment profiles

Segment	Customers	Share	Avg tenure	Avg monthly	Churn rate
Long-Tenure Value	114	22.8%	39.54	97.83	8.8%
Developing / Mixed	249	49.8%	35.7	108.14	10.0%
Newer High-Charge	137	27.4%	35.54	136.77	13.1%

Recommended actions

Established High-Value

Use proactive retention, loyalty recognition and relevant upgrades; avoid blanket discounts.

Newer High-Charge

Strengthen onboarding, explain plan value and investigate early billing or service friction.

Long-Tenure Value

Recognize loyalty, test suitable bundles and protect margins.

Developing / Mixed

Use measurable experiments and refine segments with additional customer behavior data.

Measuring impact

Run controlled A/B tests and track churn, save rate, conversion, incremental revenue, gross margin, campaign cost and complaints. Calculate net impact against a credible control group. Do not claim revenue uplift before measurement.

Evaluation results

Long-Tenure Value: accuracy 0.897, precision 0.000, recall 0.000, F1 0.000, ROC-AUC 0.962.

Developing / Mixed: accuracy 0.984, precision 1.000, recall 0.833, F1 0.909, ROC-AUC 1.000.

Newer High-Charge: accuracy 0.971, precision 1.000, recall 0.800, F1 0.889, ROC-AUC 0.993.

Small per-segment test sets create uncertainty. Use precision and recall together. These model outputs support, not replace, responsible business judgment. Segment names are heuristic and not causal.